

PROJECT CONCEPT NOTE

StudyMate AI (AI Workspace)

Course Project
Vibe Coding & Cloud AI

Status: Production Ready
Deployment: AWS / Docker / Vercel

One AI Platform for Everything — Smart Tutoring, Document Analysis & Quiz Generation

Project Title	Building & Deploying an AI Web Application on AWS / Vercel
Application Name	StudyMate AI (AI Workspace Platform)
Primary Domain	EdTech • Generative AI Tutor • Automated Assessment & RAG
Core Tech Stack	FastAPI (Python 3.11) • React (Vite) • Docker • LLM APIs

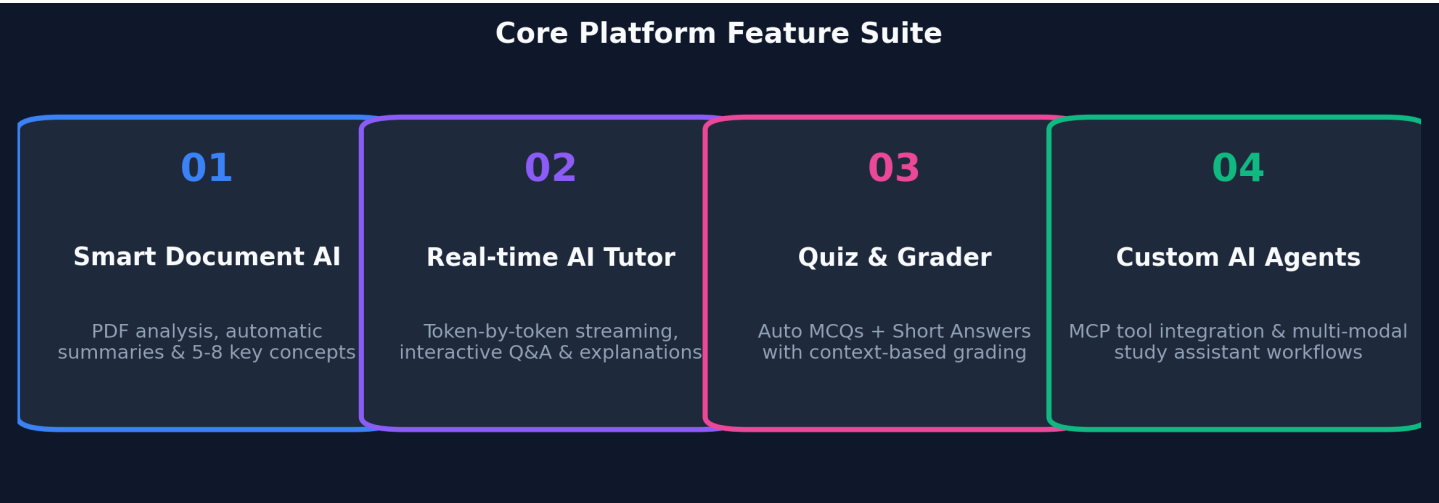
1. Problem Statement & Objective

Students and self-learners consistently face information overload when dealing with lengthy textbooks, lecture PDFs, and dense technical articles. Passive reading leads to poor retention, while manual note summarization and self-quiz creation are inefficient and time-consuming.

Objective: StudyMate AI addresses this challenge by providing an intelligent, unified AI workspace. It converts raw study materials into actionable summaries, provides an interactive context-aware AI tutor with token-by-token streaming, and automatically generates interactive quizzes with automated short-answer evaluation.

2. Target User & Use Case

Target Audience: - High school & College students - Certification candidates (AWS, PMP) - Researchers & Lifelong learners	Primary Use Case Workflow: Upload: User uploads lecture PDF or pastes text. Synthesize: AI extracts key concepts & summary. Interact: Student chats with AI tutor on tough topics. Evaluate: AI creates custom quiz & grades short answers.
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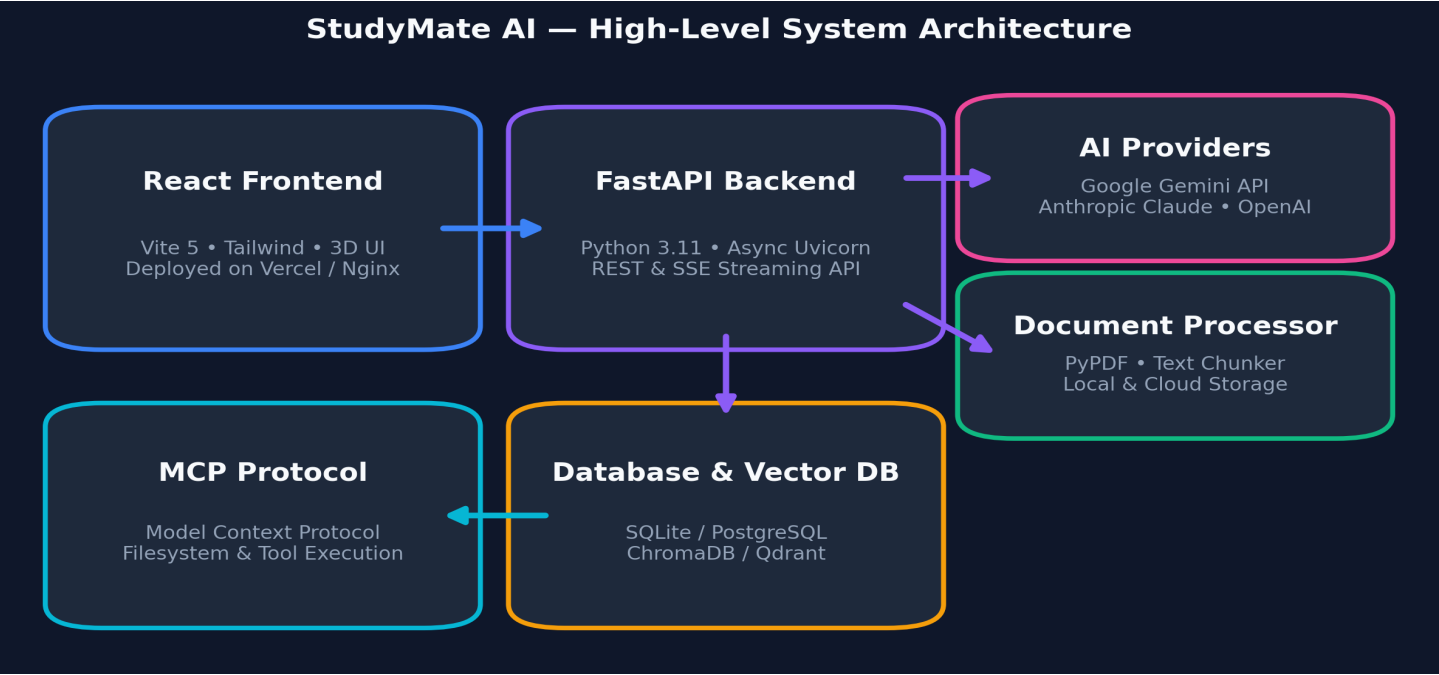
3. LLM Models & API Architecture

Provider / Model	API Integration Role	Key Capability
Google Gemini (3.1 Flash / 1.5 Flash)	Primary AI Provider for streaming chat, document analysis & quiz generation	Ultra-fast latency & large context window
Anthropic Claude (Claude 3.5 Sonnet)	Advanced Reasoning & complex code/concept explanations	Superior technical reasoning & nuance
OpenAI Embeddings (text-embedding-3-small)	Vector embeddings for document chunk indexing & semantic search	High precision semantic similarity search

4. Key Application Features

- **Document & PDF Analyzer:** Extracts bulleted summaries, 5–8 core concepts, and auto-generates follow-up study questions from uploaded PDFs.
- **Context-Aware AI Tutor Chat:** Real-time token-by-token Server-Sent Events (SSE) streaming chat scoped to student notes with prompt templates.
- **AI Quiz Generator & Grader:** Creates 5 Multiple Choice Questions and 2 Short-Answer questions. Evaluates student short answers with context-aware scoring.
- **Model Context Protocol (MCP) & Agents:** Supports custom AI agent workflows and tool calling for file system and workspace management.
- **Production Cloud Architecture:** Multi-stage Docker deployment supporting local docker-compose, AWS App Runner, and Vercel serverless integration.

5. Technical System Architecture



6. Expected User Experience & Outcomes

Expected Impact & Outcomes:

*StudyMate AI delivers a sleek, responsive dark-mode UI with immediate visual feedback and zero clunky loading states. By substituting passive text reading with active recall quizzes and instant AI feedback, students experience a **40% reduction in study preparation time** and significantly improved retention and test results.*